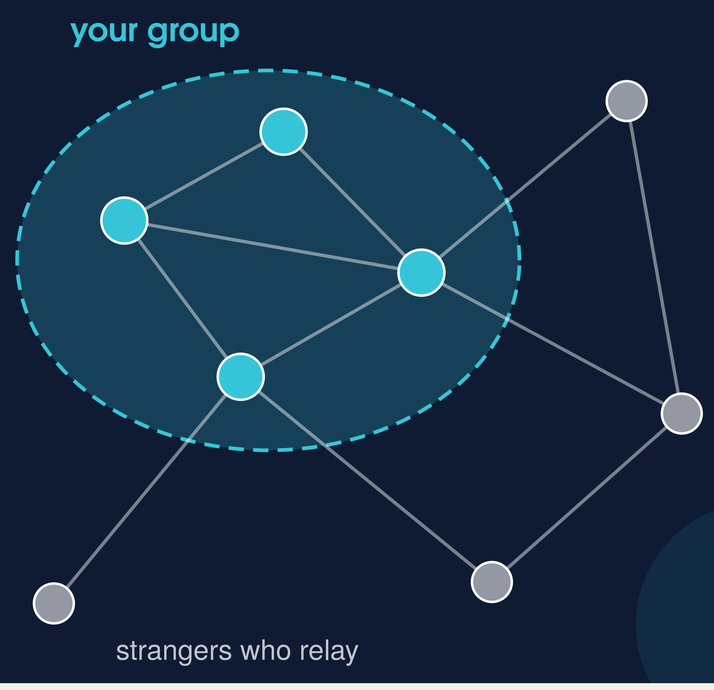
BUBBLE-BLUE

A private conversation, carried by strangers.

A Bluetooth mesh that any board can relay, and a group that only its members can read.

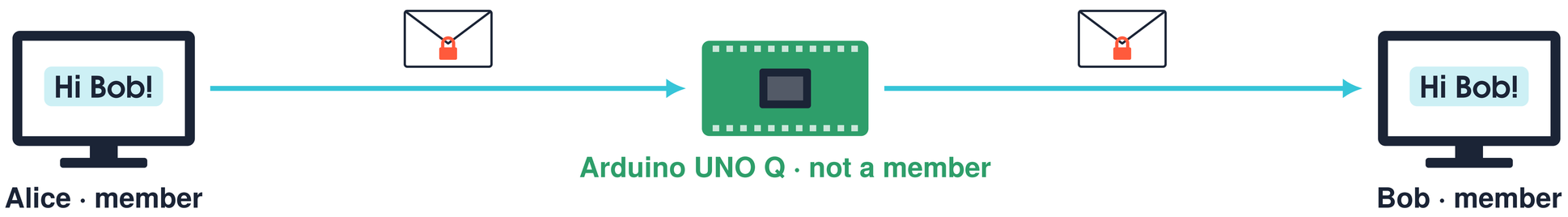
Paul-Maxence Baraton | Nadjib Achir | Philippe Jacquet INRIA Wilfred Doré Qualcomm



The demo shows messages appearing on screens. Here is what happens in between.

1 The relay carries a letter it cannot open

Between two screens, a message can pass through boards that are not in your group. They forward it anyway, still sealed. Nobody had to pair anything.



The relay reads: who sent it, where it goes, hops left

The relay cannot read: the words

2 What a spy's radio catches

A stranger listening nearby

from: Alice | to: everyone | hops left: 9

```
e4 7a 91 0c 5b ff 2d 83 c7
```

A member of the group

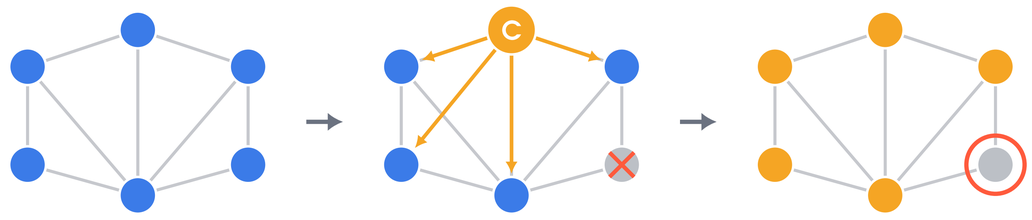
from: Alice | to: everyone | hops left: 9

Meet at the Qualcomm booth, 3 pm

Anyone can see that boards talk. Only members read what they say.

3 Removing a member

Like changing the locks: everyone gets a new key, except the one who has to go.



Everyone shares one key.

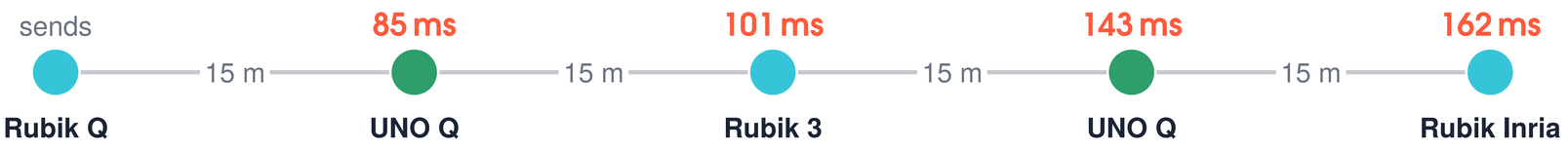
The coordinator hands out a new key, skipping one board.

The excluded board still relays. It can no longer read.

old key · new key · excluded

4 Across the building, on battery

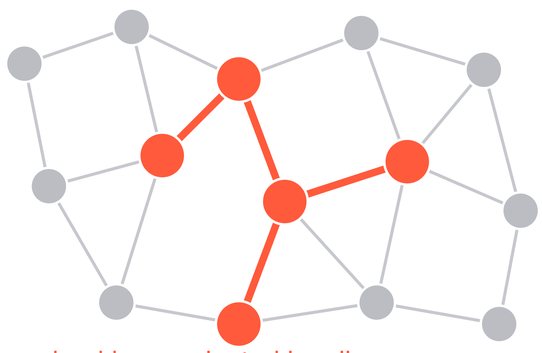
measured on the boards themselves



76/79 delivered four hops away.
Last link walked away: held to **30 m**.
One relay unplugged: **95 s** dark.
A line has no detour.

5 Who relays? The mesh decides

Each board looks at its neighbours. If they can already reach each other, it stays quiet. The boards that remain form a backbone: fewer transmissions, the same reach.



backbone, elected locally

Measured on our stand

75/75
messages reached every other board

4.4 s
from mesh start-up to reading the group

130 s
a relay whose radio was cut reads again, no reboot

0
Bluetooth pairings needed

Not claimed yet

A full day of operation, and how far every hop reaches before it breaks: only the last hop was pushed until it dropped.

Hardware

Qualcomm RUBIK Pi 3 screens · Arduino UNO Q relays
BLE L2CAP · AES-256-GCM · no pairing

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